Search Suggestions System (medium)

Problem Statement

Given a list of distinct strings `products` and a string `searchWord`.

Determine a set of `product suggestions` after each character of the search word is typed. Every time a character is typed, return a `list` containing up to three product names from the `products` list that have the same prefix as the typed string.

If there are more than 3 matching products, return 3 lexicographically smallest products. These product names should be returned in lexicographical (alphabetical) order.

Examples

1. **Example 1**:
   - **Input**: Products: ["apple", "apricot", "application"], searchWord: "app"
   - **Expected Output**: [["apple", "apricot", "application"], ["apple", "apricot", "application"], ["apple", "application"]]
   - **Justification**: For the perfix 'a', "apple", "apricot", and "application" match. For the prefix 'ap', "apple", "apricot", and "application" match. For the prefix 'app', "apple", and "application" match
2. **Example 2**:
   - **Input**: Products: ["king", "kingdom", "kit"], searchWord: "ki"
   - **Expected Output**: [["king", "kingdom", "kit"], ["king", "kingdom", "kit"]]
   - **Justification**: All products starting with "k" are "king", "kingdom", and "kit". The list remains the same for the 'ki' prefix.
3. **Example 3**:
   - **Input**: Products: ["fantasy", "fast", "festival"], searchWord: "farm"
   - **Expected Output**: [["fantasy", "fast", "festival"], ["fantasy", "fast"], [], []]
   - **Justification**: Initially, "fantasy", "fast", and "festival" match 'f'. Moving to 'fa', only "fantasy" and "fast" match. No product matches with "far", and "farm".

Try it yourself

Try solving this question here:

Java

```java
import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;

// class TrieNode {
//     TrieNode[] children = new TrieNode[26];
//     boolean isEnd = false;
// }

class Solution {
    public List<List<String>> suggestedProducts(String[] products, String searchWord) {
        List<List<String>> result = new ArrayList<>();
        // ToDo: Write Your Code Here
        return result;
    }
}
```

Test Case | Run Code Result | Stdout

```
["mobile","mouse","moneypot","monitor","mousepad"]
"mouse"
```

Use Example Testcases | Run | Submit